

Adarsh Kuthuru

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PROFESSIONAL SUMMARY

Seasoned AI/ML Professional with 8+ years of experience driving high-impact initiatives & complex problem-solving, delivering measurable revenue lifts, risk reductions, and process efficiencies through Gen AI/ML innovation, and cross-functional team leadership.

TECHNICAL SKILLS

Programming: Python, SQL, R, SAS, MATLAB, Shell, HTML/CSS, Swift, TeX, VBA, TypeScript, JavaScript

Tools: AWS/GCP suite, Git, Tableau, Snowflake, Databricks, Spark, Docker, Streamlit, Django, Flask, REST APIs, MS Office

ML/DL & Gen AI: TensorFlow/PyTorch, scikit-learn, RAG, AI Agents, Prompt & Context Engineering, MCP servers

Concepts: Deep Learning, Machine Learning (Supervised, Unsupervised, Reinforcement Learning), Statistical Inference, Optimization

Certifications: [Deep Learning](#) (Deeplearning.ai), [Machine Learning](#) (Stanford University), [Agentic Development](#) (Google)

WORK EXPERIENCE

Capital One (contractor through Zenon LLC)

Aug 2023 - present

Lead Data Scientist, Card (Apr 2025 - present); Senior Data Scientist, Card (Aug 2023 - Mar 2025)

Boston, MA

- Boosted marketing campaigns' effectiveness for a \$157B credit card portfolio by developing XGBoost models to predict spend and response rates, and strategically targeting high-value prospects, leading to a 7% increase in accounts booked and revenue annually.
- Architected a multi-agentic Gen AI system for the enrichment of stale firmographic and contact data using Google's ADK, enabling precise campaign segmentation, increasing conversion rates by 5%, and reducing marketing effort wastage by 70%.
- Owned the end-to-end RAG experimentation lifecycle - from dataset curation, chunking to reranking, retrieval evaluation - leveraging parallel hyperparameter optimization to achieve 30% accuracy gains while reducing tuning cycles from days to few hours.
- Designed and executed portfolio-wide (10M+ active cardholders) A/B testing experiments to optimize personalized credit limit upgrades and rewards strategies, identifying 15% lift in monthly customer spend.

Discover Financial Services (contractor through Zenon LLC)

Dec 2021 - Jul 2023

Senior Data Scientist, Loans (Nov 2022 - Jul 2023); Data Scientist, Loans (Dec 2021 - Oct 2022)

Boston, MA

- Spearheaded the development of an automated Gen-AI based underwriting software in Python that analyzes application data, agency reports, and triggers GPT-generated rules for optimized process progression, resulting in a 60% reduction in lead time.
- Led cross-functional initiative to upgrade legacy segmentation process for 50M+ prospects, eliminating dependencies and errors, modifying filters to reduce portfolio risk by 25%, and reducing process runtime using PySpark/Dask from 2 weeks to <2 hours.
- Drove transformative improvements in cross-selling through advanced analytics and leveraging predictive models, delivering 30% increase in application volume and 15% improvement in customer lifetime value annually.
- Designed and executed Bayesian experiments on the \$128B loan portfolio to identify and filter out high-risk applicants, thus reducing default rates by 10% while increasing approval efficiency.

Carroll School of Management, Boston College

Aug 2019 - Jun 2021

Graduate (PhD) Researcher

Boston, MA

- Built clustering-based orthogonalized scores to predict fund manager skill, analyzed its relationship with their performance using multivariate panel data regression, and designed a trading strategy based on it, securing a statistically significant 25% ROI annually.
- Modeled the causal relationship between NLP-based sentiment metrics from 300+ media sources and stock returns to develop predictive buy/sell indicators for portfolio management; achieved a statistically significant portfolio returns of 15% per annum.
- Designed and executed regression discontinuity and instrumental variable regressions, to evaluate the impact of activist investments on firm's market value, and built an investment strategy around it resulting in a 10% increase in ROI annually.

Center for Analytical Finance, Indian School of Business

Jun 2016 - Jul 2019

Senior Research Associate, Data Science (Jul 2017 - Jun 2019); Research Associate, Data Science (Jun 2016 - Jun 2017)

India

- Leveraged time-series models and genetic algorithms to study risk patterns in global equity markets and develop investment strategy that optimizes returns and risks, delivering an average annual risk-adjusted return of 20%.
- Processed 7B+ patents/citations from USPTO to create innovation efficiency metrics; analyzed their relationships with equity returns using econometric modeling techniques, built a portfolio based on it generating a 15% ROI annually.
- Built logistic regression models to estimate firm-level credit risks and mitigated exposure to high-risk firms thus minimizing expected annual losses in investment portfolios by 30%.
- Organized statistical inference workshops and provided mentorship to new hires and interns. Excelled in high-pressure environments, managing heavy workloads and meeting tight deadlines, resulting in a 100% rating in annual reviews.

EDUCATION

Indian Institute of Technology (IIT), Kharagpur, India

Jun 2011 - May 2016

Bachelor of Technology & Master of Technology in Engineering